AWG19 POLARIZATION DRIVER — CONNECTED SCHEMATIC

External coil: 11.430016 ohm / 25.92669 mH
Polarize: Q1+Q4 on. Q5 off.
Discharge: Q1+Q4 off, clamp conducts.
Receive: Q5 on, U2A on, other U2 switches off.

Q4 / Q5 need MCU dead time.
No hardware interlock yet.
Regulated 12 V supply required.

LOCAL BYPASS: place each capacitor at its named IC supply pin

Sensor: 1.65 V at zero current; 1 V/A.
3.3 V MCU ADC assumed.

DRAFT — component ratings / footprints still need review.
Current monitor measures polarization, not clamp current.
Kelvin route U4 input pair to R9 pads.
Signal and MCU labels join only the named remote nets.

U3 LTC1693-5
U1 LTC1693-1
J1 REGULATED SUPPLIES
J2 EXTERNAL AWG19
J3 MCU CONTROL / ADC
J4 TO AMPLIFIER
Q1 IRF530
Q4 IRF9640
Q5 IRF530
U2A U2B U2C U2D U2E ADG5412F
U4 INA240A1PW
R1 22
R2 22
R3 2.2
R4 100k
R5 100k
R6 100k
R7 10k
R8 10k
R9 0.05 / 0.5W
R10 100
C1 1u
C2 10n
C3 1u U1–8
C4 1u U1–6
C5 1u U3
C6 100n U2
C7 100n U2
C8 100n U4
CLAMP_MID
FAST – TBD2 TVS 43V – TBD

ECE 455 — Jun Yang	
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